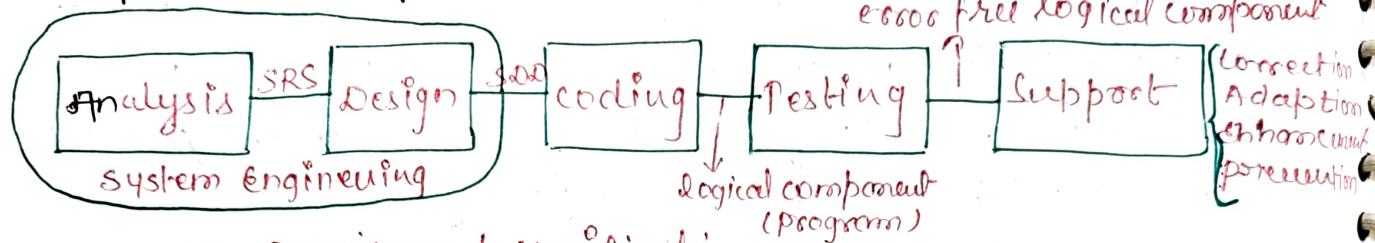


Conventional process model :-

- This models are used to develop the final product at a time after a long schedule.
- Different frameworks are employed in this category:-

① Waterfall model :-

- The framework of the waterfall model is



SRS → S/W Requirement Specification

SDD → S/W Design description Document

System Engineering:-

- S/W is a subset of the system, so there is a need of analyzing the existing system to finalize the objective or goal of the product.

Analysis :-

- Based on the objective of the S/W diff. requirements are finalized.
- The requirements are classified into functional & non-functional requirements.
- The outcomes of the analysis stage is documented into the SRS document.

∴ SRS doc. consists

- Goal of the sys.
- Functional requirements of the sys.
- Non-functional " " " "

Design :-

- In this stage 4 levels of the design is performed to translate the generalized problem statement into procedural description (algorithms)

→ 4 levels of designs are

- 1) Data level: create the Data structure.
- 2) Arch: " : " " Hierarchical "
- 3) Interface " : defines the logical relationships b/w modules.
- 4) component " : translates the general stat. into ~~pro~~ technical description.

→ the outcomes of the design stage is documented into the s/w design description document.

Coding:-

→ In this stage, the procedural description is translated into m/cn. readable format by using the appropriate background lang.

Testing:-

→ In this stage, various test plans are implemented on the developed prog. to cover the functional & structural errors. the objective of the testing is to prove the system what extent it becomes failure.

After the testing the error free component is deployed into the customer's place.

Support:-

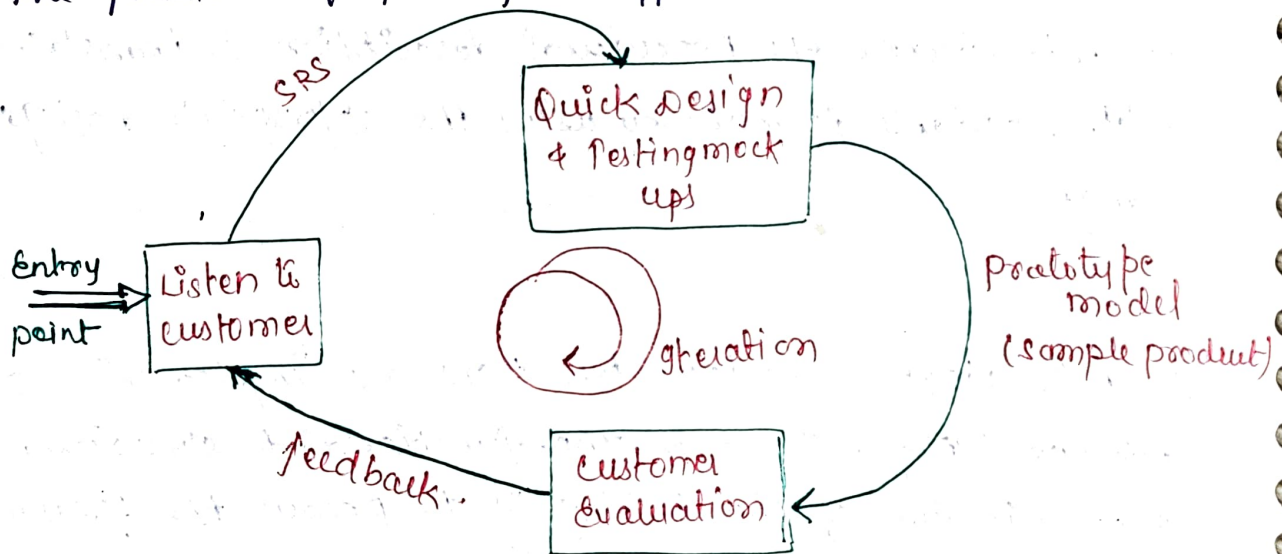
when the developed product is present in the operational state then support stage is required to maintain the product from the various side effects.

li:- waterfall model is suitable when the requirements are more clear & the customer patience level are very high. becoz in this model the final product is available after a long schedule.

- This model is not suitable to design the complex products bcoz it is not possible to describe complete requirements at the early stages of development.
- In this model testing stage takes more effort bcoz the complete system is undergoes testing at a time.

② prototype model :- (elicitation model) (gather)

- when the requirements are not clear then prototype model is used to finalize the SRS doc.
- the framework of the prototype model is



- In this framework, the customer involvement is present to finalize the requirements.
- At the initial stage min. set of the requirements are documented.
- With those requirements quick design practices are used to develop the sample product.
- The sample product is deployed into the customer place.
- After using the prototype product the customer generates the feedback to the development team.
- When the feedback form contains the new req., then the SRS doc. is modified & the same process will be continued.

- when the feedback doesn't contain the new requirements then the corresponding sample product's SRS doc will be finalized.
- The prototype model itself a "Elicitation" technique.
- After finalizing the SRS doc, two types of the prototype model used to develop the final product namely:

① open-end prototype

② close - " "

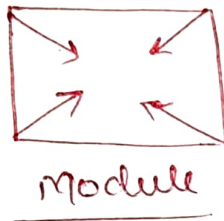
- After finalizing the SRS document use the same framework to develop the final product is called as open-end prototype.
- After finalizing the SRS doc. using the another framework to develop the ~~another~~ ^{final} product is called as close-end prototype.

③ RAID MODEL :-

- RAID model is used to develop the product when the requirements are more clear & the schedule is very short.
- To develop the project in the limited time span when the requirements are clear then use the modularity concept.
- Modularity means divide the project into ^{independent} submodules. develop them simultaneously later integrate them to produce the final project.
- Before approach to the modularity there is a need of considering the man-power requirement.
- when the orgⁿ. is having the efficient man-power then divide the man-power into teams.
- The independent modules are assign to the teams to develop them simultaneously.
- modularity concept is associated with the two factors

① cohesion ② coupling

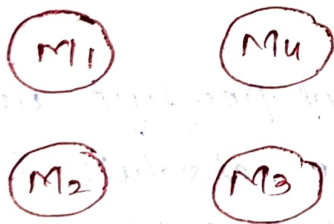
- cohesion is a quantitative measure of what extent the module is independent from the other module.
- The cohesion of the module is denoted as



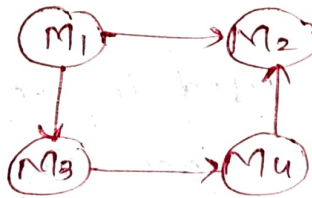
- cohesion indicates the functional strength of the module

- Coupling is a quantitative measure of what extent the module is independent on other module.

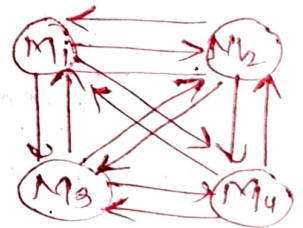
- Different levels of coupling denoted as



no coupling



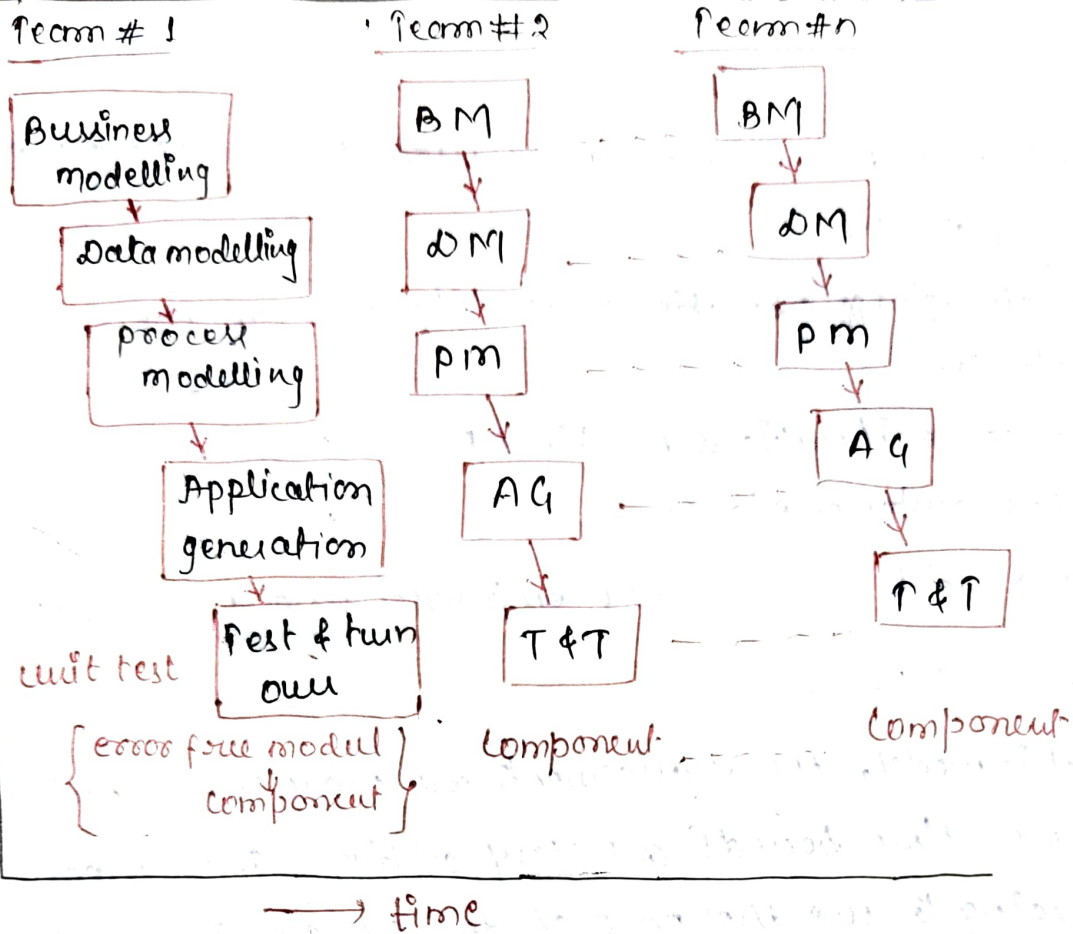
loosely coupling



Tightly coupling

- the effective modularity maintains the high ^{cohesion} ~~collaboration~~ & low coupling.

- modularity concept requires the integration process.



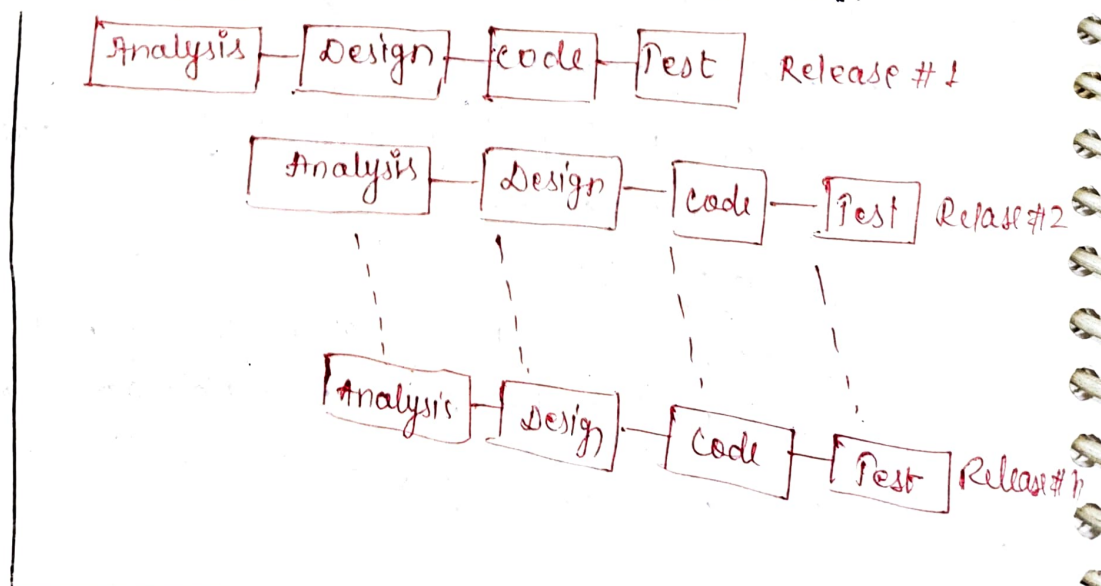
- In Business modelling, existing system is analyzed to finalize the business functⁿ.
- Based on the objective of the business various data objects are gathered during the Data modelling stage.
- In the process modelling info. flow is created to satisfy the business functⁿ. either by adding or deleting or modifying the data objects.
- In the applⁿ generation stage the info. flow is translated into mach. readable format.
- During the testing stage diff. test cases are implemented to cover the errors.
At the end of testing we have a error free component.
- In the RAD model testing stage takes less effort bcoz only the part of the prog. is undergoes S.P.
- In the RAD model integration cost is included.

Evolutionary process model :-

- Evolutionary process models are used to develop the product part by part.
- In each release operational quality part of the system is available.
- After some iterations the final product will be available in the customer place.
- There are diff. frameworks employed under this category :-

① Incremental model :-

- In this model, the requirements are more clear but the customer time bounds are very critical. That means customer is going to use the part of the system in a short time span.
- Incremental model uses the classic life cycle framework to develop the product Release by Release.
- In each Release additional functionality is added to the existing system, so after some iteration the final ~~proje~~ product will be available to the customer place.
- The framework of the incremental model is :-



Spiral model :-

→ This framework consists 6 activities name as :-

- ① Customer Commⁿ.
- ② Planning
- ③ Risk Analysis
- ④ Engineering
- ⑤ Construction & Release
- ⑥ Customer evaluation.

① Customer Commⁿ!:

→ In this framework customer involvement is present to finalize the Requirements.

→ After finalizing the requirements all are documented into the SRS.

② Planning!:

→ Planning means estimation.

→ In the planning stage various attributes of the s/w is undergoes estimation i.e size of the s/w, effort of the s/w, schedule & cost of the s/w.

③ Risk Analysis!:

→ In this stage, Risk impact is estimated. This value will be added to the project Budget. So, during the project development if any risk condⁿ is present then s/w Budget is not extended.

→ It means, on time & within the budget the s/w is developed when the Risk estimation is maintained in the plan.

④ Engineering:

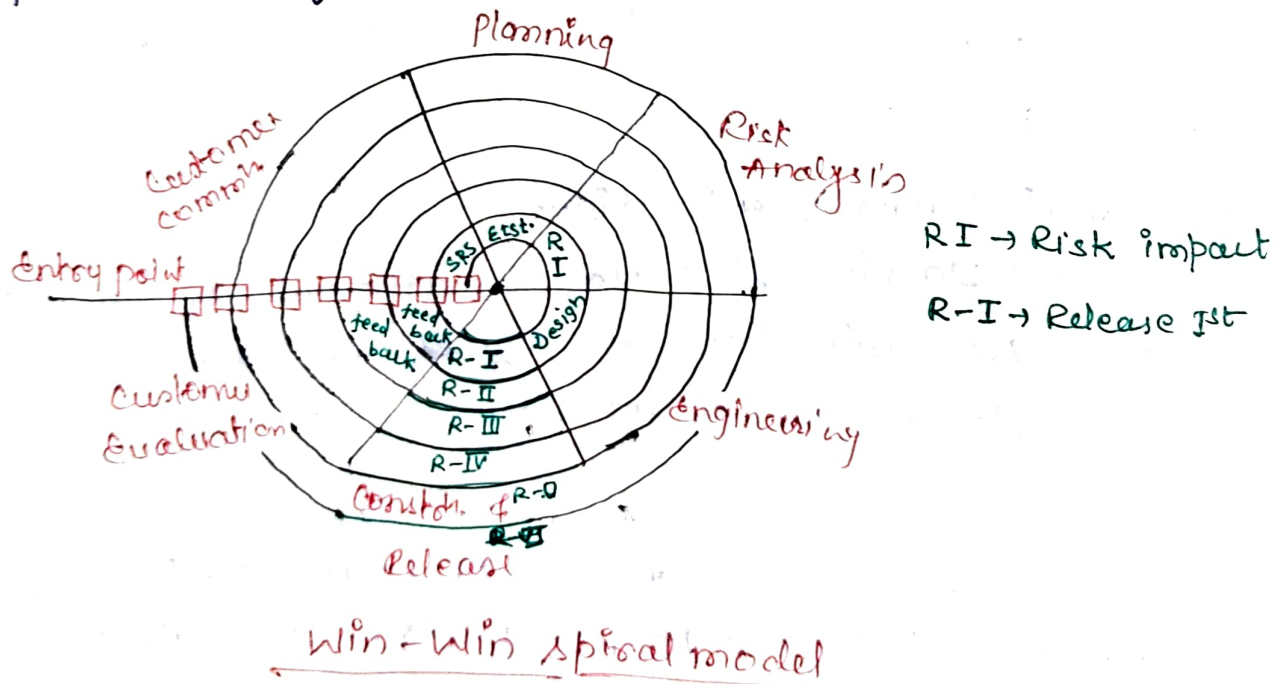
- By using the software engineer practices design document is finalized in this stage.

⑤ Construct. & Release:

- Coding & Testing activities are performed in this stage.
- After the testing the product is deployed into the customer's site.

⑥ Customer Evaluation:

- In this framework, continuous qualitative feedback is accepted from the customer to improve the product functionality.
- ∴ this model is a continuous improvement process model.
- the framework of this model is



- In this model the base product contains the limited set of functionalities.
- By taking the continuous improvement process the product functionality will be increased.
- so the customer satisfaction is very high simultaneously developer's skill level is also elaborated.
- ∴ it is also called as the win-win spiral model.

④ Engineering:

→ By using the software engineer practices design document is finalized in this stage.

⑤ Construct. & Release:

→ Coding & Testing activities are performed in this stage.

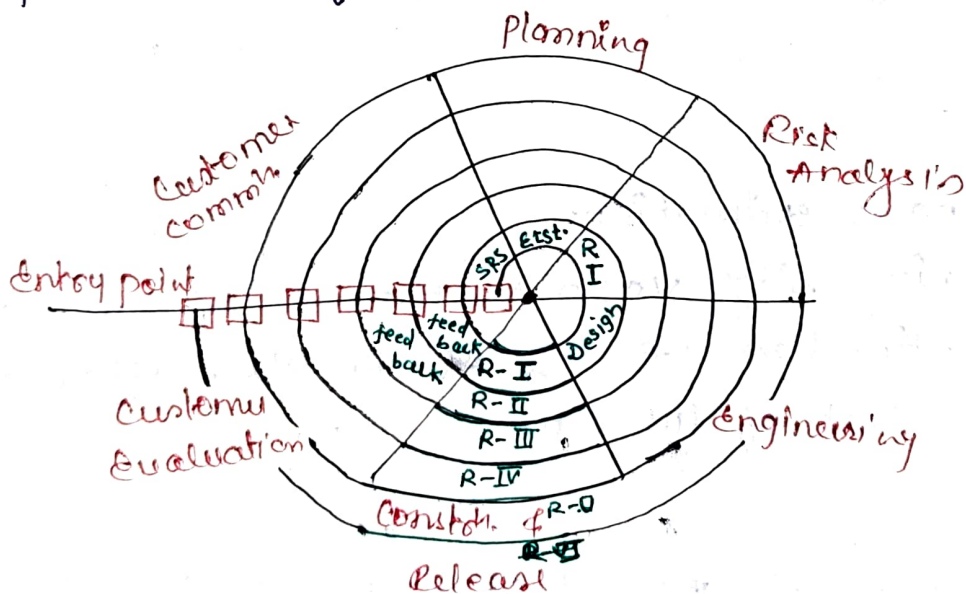
→ After the testing the product is deployed into the customer's site.

⑥ Customer Evaluation:

→ In this framework, continuous qualitative feedback is accepted from the customer to improve the product functionality.

∴ this model is a continuous improvement process model.

→ the framework of this model is



RI → Risk impact

R-I → Release 1st

→ In this model the base product contains the limited set of functionalities.

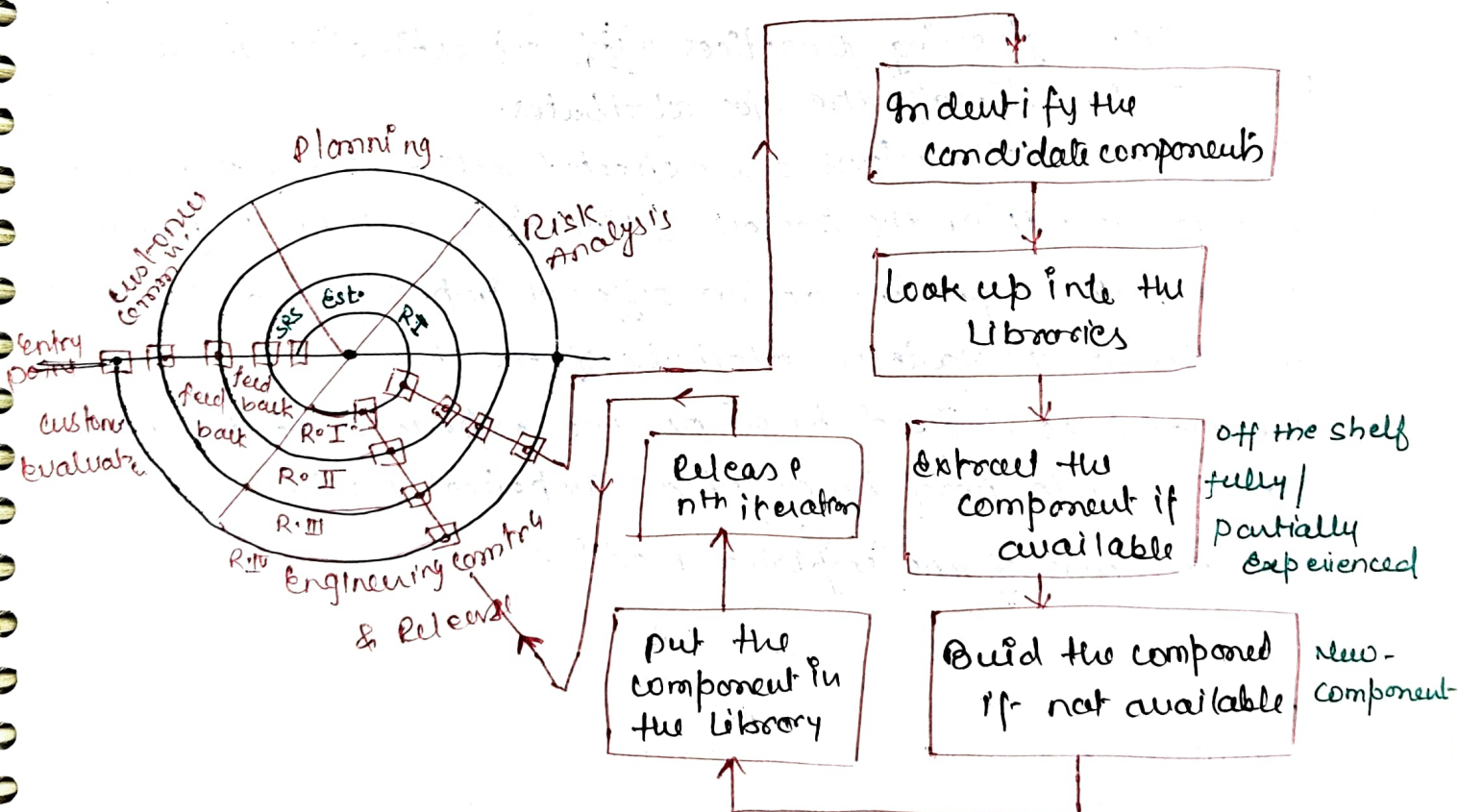
→ By taking the continuous improvement process the product functionality will be increased.

so the customer satisfaction is very high simultaneously developer's skill level is also elaborated.

∴ it is also called as the win-win spiral model.

③ Component Based process model :-

- when the components are used in the project then the project development effort will be minimized.
- This framework still uses the spiral model approach but before implementing the design document components are extracted into the libraries.
- Due to the systematic estimated plan this process model is also comes under the quantitatively & qualitatively controlled process models.
- the framework of the CB Life cycle is :-



- After finalizing the design document the developer can able to identify the no. of the modules present in the project.
- Before developing the project verify the libraries for the components.
- If the component is available directly use them in the project.
- If not available develop the functionality from the base line.
- Before deploying the product into the customer place put the

developed components into the library.

→ They may become the components for the future projects.

→ In this framework, also continuous qualitatively feedback is accepted from the customer to improve the product functionality.

→ In this framework the project is developed on time & within the budget.